Individual Assignment

Binary Logistic Regression Interactive Dashboard

Module Code	ST 4035
Module Name	Data Science
Assignment Type	Individual Assignment
Due Date	20 May 2026 | 11:59 PM
Submission	LMS upload + GitHub Pages link

Academic Year 2025/2026 – Semester II

Assignment Overview

This assignment requires you to design and deploy a fully interactive web-based dashboard that demonstrates key concepts in binary logistic regression, including class imbalance handling and the effect of different link functions. The dashboard must be built using Python or HTML, CSS, and JavaScript, published on GitHub Pages, and documented in a brief report uploaded to the LMS.

Your dashboard must run entirely in a web browser without any server-side code. All interactivity must update charts and metrics in real time.

Required Controls

1. Sample size slider — allow the user to vary n from 100 to 2,000 in steps of 50. Display the current value (e.g. n = 750) next to the slider. All charts must update immediately when the slider moves.
2. Class imbalance ratio slider — vary the minority-class proportion from 5% to 50% in steps of 5. Display the selected percentage next to the slider.
3. Balancing method selector — provide clearly labelled toggle buttons or radio buttons for four options: None (imbalanced), Oversampling, Undersampling, SMOTE. Only one option may be active at a time.
4. Link function selector — provide toggle buttons for three options: Logit, Probit, Complementary log-log. Only one option may be active at a time. The active link function must be visually highlighted.

Required Metrics Panel

Display at minimum the following four live-updating metric cards, clearly labelled:

- Total samples (reflecting the current balancing method and sample size).
- Class 1 count (minority class).
- Class 0 count (majority class).
- Imbalance ratio (formatted as 1 : x.x).

Required Charts

All four charts below are mandatory and must update whenever any control changes:

Chart 1 — Class Distribution (Before and After Balancing)

A grouped bar chart showing the Class 1 and Class 0 counts side-by-side for both the original (imbalanced) dataset and the balanced dataset under the selected method. Use distinct, accessible colours for the two classes. Include a legend.

> 💡 *For SMOTE, the synthesised minority count should approach the majority count (e.g. 85% of majority). Show this clearly in the chart.*

Chart 2 — Fitted Probability Curves (Link Functions)

A line chart plotting P(Y = 1 | eta) vs. the linear predictor eta over the range [-4, 4] for all three link functions simultaneously. The currently selected link function must be drawn with a thicker or bolder line. Use distinct line styles (solid, dashed, dotted) in addition to colour to differentiate the three curves.

Chart 3 — ROC Curve

A plot of Sensitivity (True Positive Rate) vs. 1 - Specificity (False Positive Rate) for the model fitted using the selected link function, sample size, and balancing method. Include a diagonal reference line representing a random classifier. Display the AUC value in the chart area or in the metrics panel.

Chart 4 — Confusion Matrix (Normalised)

A visualisation (bar chart or heatmap) of the normalised confusion matrix showing True Positive rate, False Negative rate, True Negative rate, and False Positive rate. Values must be proportions in [0, 1], not raw counts.

Information Footer

Below the charts, display a summary row or panel showing the current values of: AUC, Accuracy, Sensitivity, Specificity, active link function, and active balancing method. This information must update with every control change.

Technical Requirements

- Single HTML file (index.html) or use Python, R, or server-side code.
- All controls must trigger real-time updates without page reloads.
- The dashboard must be fully functional on Chrome and Firefox.
- Responsive layout — the dashboard should be usable on a 1280px wide screen.
- Accessible colours — do not rely on colour alone to distinguish data series; use line styles, patterns, or labels.

GitHub Pages Deployment

Your dashboard must be publicly accessible via GitHub Pages. Follow these steps:

Step 1 — Create a GitHub repository

5. Sign in to github.com. Create a new public repository named ST4035-dashboard (or similar).
6. Do not initialise with a README if you plan to push from your local machine.

Step 2 — Add your file

7. Place your completed index.html in the root of the repository.
8. Commit and push to the main branch.

Step 3 — Enable GitHub Pages

9. Go to your repository on GitHub. Click Settings > Pages.
10. Under Source, select Deploy from a branch. Choose main branch, / (root) folder. Click Save.
11. After a minute or two, your dashboard will be live at: https://yourusername.github.io/ST4035-dashboard

Step 4 — Verify deployment

12. Open the published URL in a browser. Confirm that all charts render and all controls work correctly.
13. Copy the live URL — you will include it in your report and share it on the WhatsApp group.

Brief Report (LMS Submission)

You must upload a PDF or Word report (maximum 3 pages) to the LMS by the deadline. The report must contain the following sections, in order:

WhatsApp Group Sharing

As a final step, post the following to the ST 4035 WhatsApp group before the deadline:

- A screenshot of your completed dashboard (the same screenshot included in your report is acceptable).
- Your GitHub Pages URL as a clickable link.
- Your name and index number in the message.

Academic Integrity

All submitted work must be your own. You may consult online documentation, tutorials, and references. However:

- You must not copy another student's code or report.
- You must not submit work generated entirely by an AI tool as your own without understanding or modification.
- If you use any online resource or tool, cite it in your report.
- All students are expected to be able to explain and demonstrate their dashboard in person if requested by the examiner.

Violations of academic integrity will be handled under the University's Academic Misconduct Policy.

Good luck!

Submit before 20 May 2026, 11:59 PM — Late submissions incur a 10-mark deduction per day.